

ColonyMind

A label-free visual learner that grows structure only when information earns it.



THE IDEA IN 20 SECONDS

ColonyMind begins with a small adaptive population instead of a large fixed network. Unlabeled retinal information acts as **food**. Persistent prediction error can earn new cells, organisms, and cooperation; familiar structure is consolidated as memory, so learned input should stop feeding unnecessary growth.

CELLS

ORGANISMS

MICRO LAYER

MEMORY

Important: this is bio-inspired software, not a biological simulation. The metaphors make an inspectable computational mechanism easier to reason about.

Nature as a design language

Nature analogy	Software meaning	Role in learning
Food	Unlabeled 64 x 64 retinal patches and the residual information they contain.	Novel residuals sustain adaptation; familiar input offers progressively less useful food.
Cell	A small learned adaptive unit or prototype.	Encodes a local visual regularity and contributes to an organism.
Organism	A compact, simulated local expert made from cells.	Grows and specializes where its contribution improves representation.
Micro-signature	An intermediate specialist for fine contours and local detail.	Helps separate globally similar forms, such as circles and squares.
Colony	Cooperation among organisms when shared structure is useful.	Combines complementary specialists without forcing every organism into one monolith.
Memory engram	Consolidated familiar structure.	Preserves learned organization and reduces the need to keep growing on known input.

The self-structuring loop: Retina → residual food → local specialization → cellular growth → useful cooperation → memory → less residual food on familiar input.

A demand-driven alternative to fixed capacity

Question	Conventional fixed architecture	ColonyMind hypothesis
When is capacity chosen?	Mostly before training through layer width, depth, and parameter count.	During learning, in response to persistent information residuals.
Where does specialization live?	Distributed across a predetermined topology.	In inspectable cells, organism experts, micro-signatures, colonies, and memories.
What happens when input is familiar?	The full configured network still executes.	Useful food declines; growth should stabilize and consolidated structure can be reused.
How is change studied?	Retrain or replace a model configuration.	Preserve an immutable baseline and run bounded variants in fresh isolated engines.

How computational efficiency is meant to emerge

The system follows a demand-driven capacity rule: allocate structure while unexplained information remains, reward only useful contribution, consolidate stable structure into memory, and reduce births when the stream is already digested.

CAPACITY	ADAPTATION	REUSE
Avoid reserving the final architecture in advance.	Concentrate new structure where residuals persist.	Preserve familiar structure instead of repeatedly relearning it.

THE TESTABLE HYPOTHESIS

If useful residual food declines, births and structural updates should decline with it while predictive quality remains stable. If novelty returns, growth should resume where the new residual appears. This can be compared directly with a matched fixed-capacity baseline.

What is measured today

The application records cell and organism populations, structural births, prototype updates, colony formation, memory consolidation, loss behavior, and evaluation quality. These make the growth policy inspectable and support controlled comparisons between versioned experiments.

WHAT IS NOT PROVEN YET

Current resource values are engineering **proxies**. A controlled benchmark must still measure wall-clock time, peak memory, FLOPs or accelerator utilization, and physical energy against matched fixed-network baselines before claiming real computational or energy savings.

The responsible one-line claim

ColonyMind tests whether learning systems can earn capacity from novelty, then stop growing when useful information food runs out.

One-minute interface tour

Retinal stream	A 64 x 64 unlabeled stimulus with changing scale, rotation, position, noise, occlusion, and filled or outline rendering.	This is the information food.
Digestion light	Red means abundant residual food, amber means adaptation is slowing, and green means the current stream is mostly digested.	Green is a calibrated indicator, not proof of general intelligence.
Living architecture	Drag to rotate, use the wheel or pinch to zoom, and click one organism to inspect it.	Every rendered object reads live engine state.
Selected organism	Shows cells, lineage, lifecycle, energy proxy, micro-signature affinities, colony, and memories.	Only the selected label appears, keeping the habitat readable.
Draw & Audit	Draw a circle, triangle, or square. Compare the learned community with an independent geometric auditor.	The probe is read-only and never trains the learner.
Research Auditor	GPT-5.6 interprets a frozen aggregate snapshot and exposes limitations and next experiments.	It cannot access raw pixels or modify learning.
Experiment Studio	Converts a validated proposal plus optional human instructions into a bounded protocol.	Every version gets a fresh engine; Baseline v1 stays locked.

Read the 3D visual language

• Cell / organism	Learned specialists and their branching tissue.	◇ Information food	Retinal patches moving into the learner.
○ Colony membrane	Cooperation among organisms when cohesion pays.	● Memory engram	Consolidated familiar structure and recall evidence.

A focused two-minute judge path

1. **Observe learning.** Let the retina advance. Relate incoming information food to births, growth, micro-signatures, colonies, and memories. Watch the digestion light move as residual food changes.
2. **Explore structure.** Rotate and zoom the 3D field. Select one organism and inspect its tissue, lifecycle, lineage, affinities, and colony.
3. **Challenge recognition.** Draw a shape in Draw & Audit. Check whether the learned community and external geometric auditor agree.
4. **Reveal evaluation.** Run the hidden-label read-only evaluation. Labels are never used for training; they exist only in the frozen evaluator.
5. **Ask GPT-5.6.** Click **Audit run**. Before and after state hashes must match. A fresh request may take 30 to 60 seconds; a repeated state can return from cache.
6. **Version the next question.** Add an optional constraint and create an isolated protocol. Baseline v1 remains locked and no generated code is executed.

Best viewing sequence: Retina → digestion light → 3D architecture → selected organism → Draw & Audit → evaluator → GPT-5.6 audit → versioned experiment.

How to interpret the metrics

Metric	Better direction	What it answers
Purity	Higher	Do hidden categories dominate individual learned communities? Purity alone can reward fragmentation.
NMI	Higher	How completely does the full community assignment align with hidden categories?
ARI	Higher	How well do sample pairs agree after correcting for chance?
Fragmentation	Near 1.0	Does one hidden category require many communities instead of a compact representation?
Loss trend	Lower / stable	Is recent reconstruction improving under the system's own loss definition?
Resource score	Context only	Engineering proxy for structure and activity - not watts, FLOPs, or measured physical memory.

REPRODUCIBLE BASELINE

Five seeds, 240 training steps per seed, and 72 balanced held-out samples per seed produced purity 1.000, mean NMI 0.9699, mean ARI 0.9641, and mean fragmentation 1.2. Every evaluator state hash was preserved.

SCIENTIFIC BOUNDARY

This is a synthetic circle, triangle, and square benchmark. It demonstrates a transparent self-structuring mechanism, not natural-image performance, camera generalization, or measured energy efficiency.

Why GPT-5.6 cannot alter the result

The Research Auditor receives aggregate metrics, structural counts, evaluation evidence, and declared limitations. It receives no raw retina, drawings, cell prototypes, mutable engine reference, tools, or learning endpoint. The Experiment Designer may emit only a schema-validated allowlisted protocol; each version runs in a fresh engine and the submitted baseline remains immutable.

The scientific trust chain

1. Learn	Retina enters without semantic labels.	Prevents answer leakage into training.
2. Freeze	Evaluation reads a snapshot that cannot update the learner.	Separates learning from measurement.
3. Challenge	GPT-5.6 receives aggregates and limitations through a constrained schema.	Adds critical interpretation without a write path.
4. Experiment	Approved variations run in fresh engines beside locked Baseline v1.	Makes change comparable and reversible.

Claim only what the evidence supports

SUPPORTED TODAY	FUTURE VALIDATION
Label-free organization on a controlled three-shape benchmark; inspectable structural adaptation; isolated GPT-5.6 auditing; reproducible five-seed evidence.	Natural images and camera input; unseen-category transfer; superiority over matched baselines; direct compute, memory, and physical-energy savings.

[Try the live demo](#) — [Inspect the source and evidence](#)